

QUANTITATIVE ANALYSIS FOR MANAGEMENT –II

Multiple Regression

Regression Analysis...

- Our objective is to ***analyze the relationship*** between interval variables;
- Regression analysis is used to predict the value of one variable (the ***dependent variable***) on the basis of other variables (the ***independent variables***).
- Dependent variable: denoted **Y**
- Independent variables: denoted **$X_1, X_2, \dots, X_k$**

Regression Analysis...

- If we are interested ***only*** in determining whether a relationship ***exists***, we may employ ***correlation analysis***.
- Regression analysis with ***two-variables*** is sometimes called ***simple linear regression***.
- Mathematical equations describing their relationships are also called ***models***
- 2 types: deterministic or probabilistic.

Model Types...

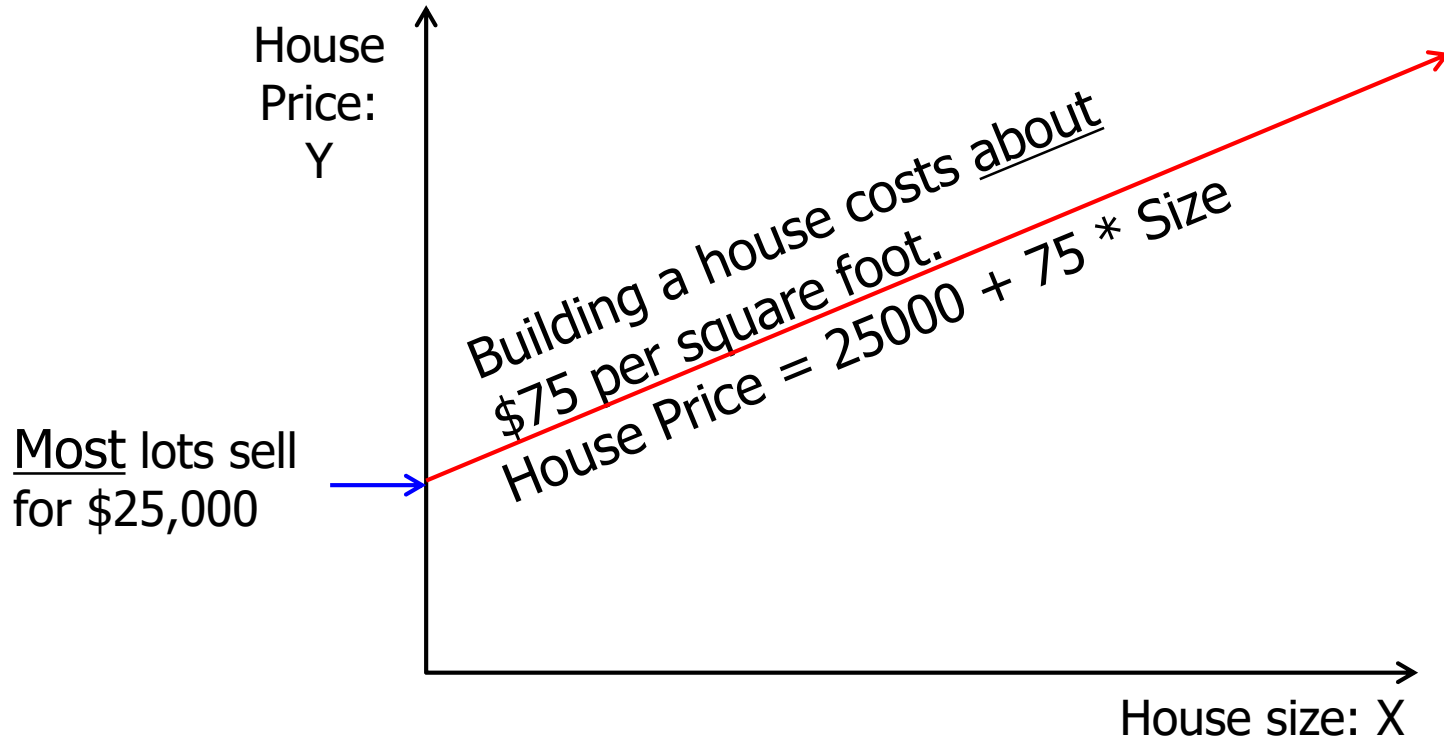
- **Deterministic Model:** an equation or set of equations that allow us to *fully determine* the value of the dependent variable from the values of the independent variables.
- Contrast this with...
- **Probabilistic Model:** a method used to capture the *randomness* that is part of a real-life process.
- E.g. do all houses of the same size (measured in square feet) sell for exactly the same price?

A Model...

- To create a probabilistic model, we start with a deterministic model that ***approximates the relationship*** we want to model and add a ***random term*** that measures the ***error*** of the deterministic component.
- Deterministic Model:
 - The cost of building a new house is about \$75 per square foot and most lots sell for about \$25,000.
 - Hence the approximate selling price (**y**) would be:
 $y = \$25,000 + (75\$/\text{ft}^2) * x$
 - (where **x** is the size of the house in square feet)

A Model...

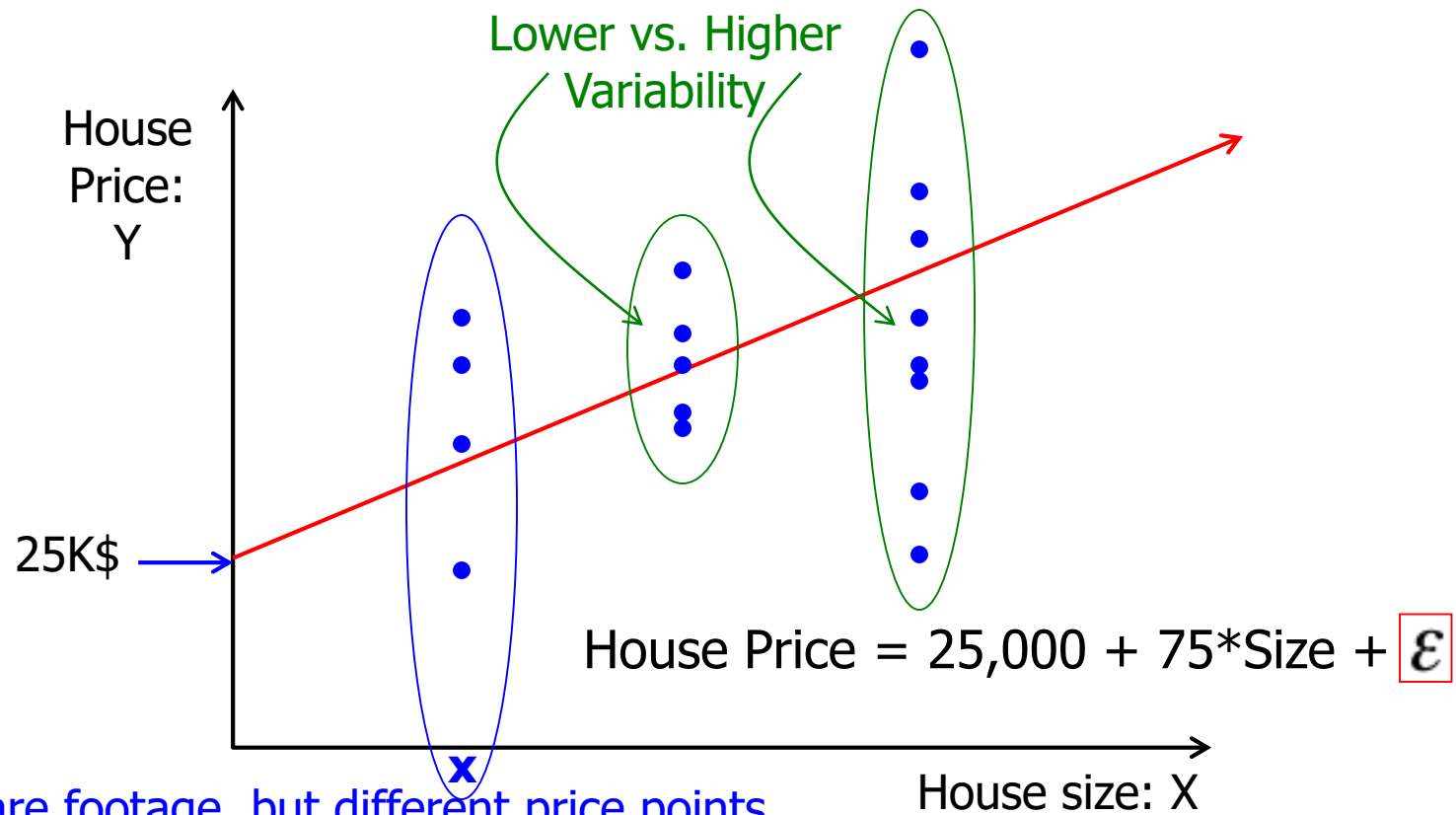
- A model of the relationship between house size (**independent variable**) and house price (**dependent variable**) would be:



In this model, the price of the house is completely **determined** by the size.

A Model...

- In real life however, the cost of houses will vary, even among for houses of the same size:



Same square footage, but different price points
(e.g. décor options, cabinet upgrades, lot location...)

Random Term...

- We now represent the price of a house as a function of its size in this Probabilistic Model:

$$y = 25,000 + 75x + \varepsilon ,$$

where ε (Greek letter epsilon) is the ***random term*** (a.k.a. ***error variable***).

- It is the difference between the ***actual*** selling price and the ***estimated*** price based on the size of the house (model based price) .
- Its value will vary from house sale to house sale, even if the square footage (i.e. x) remains the same.

Simple Linear Regression Model...

- Generally, a straight line model with one independent variable is called a ***simple linear regression model***.

$$\boxed{y} = \boxed{\beta_0} + \boxed{\beta_1} \boxed{x} + \boxed{\varepsilon}$$

dependent variable

independent variable

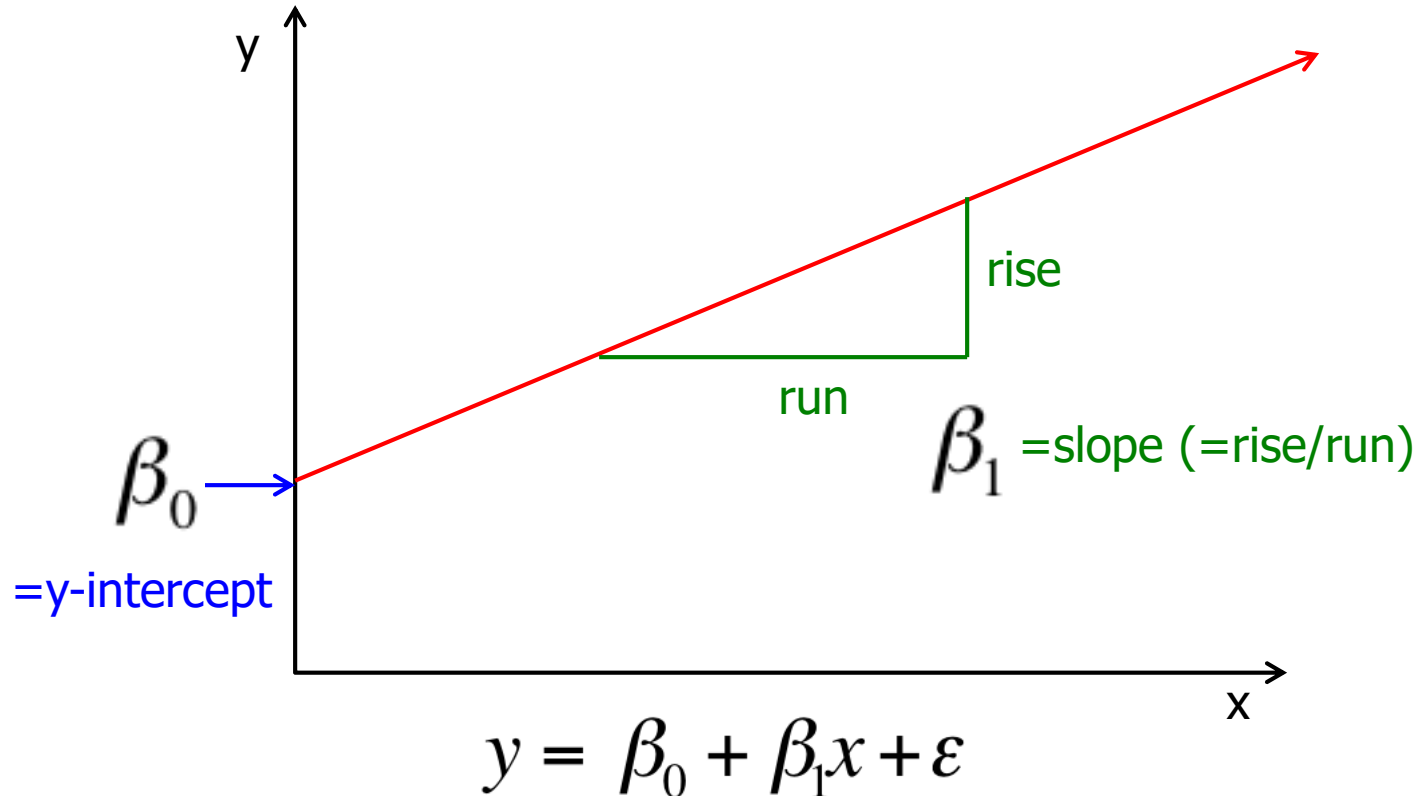
y-intercept

slope of the line

error term

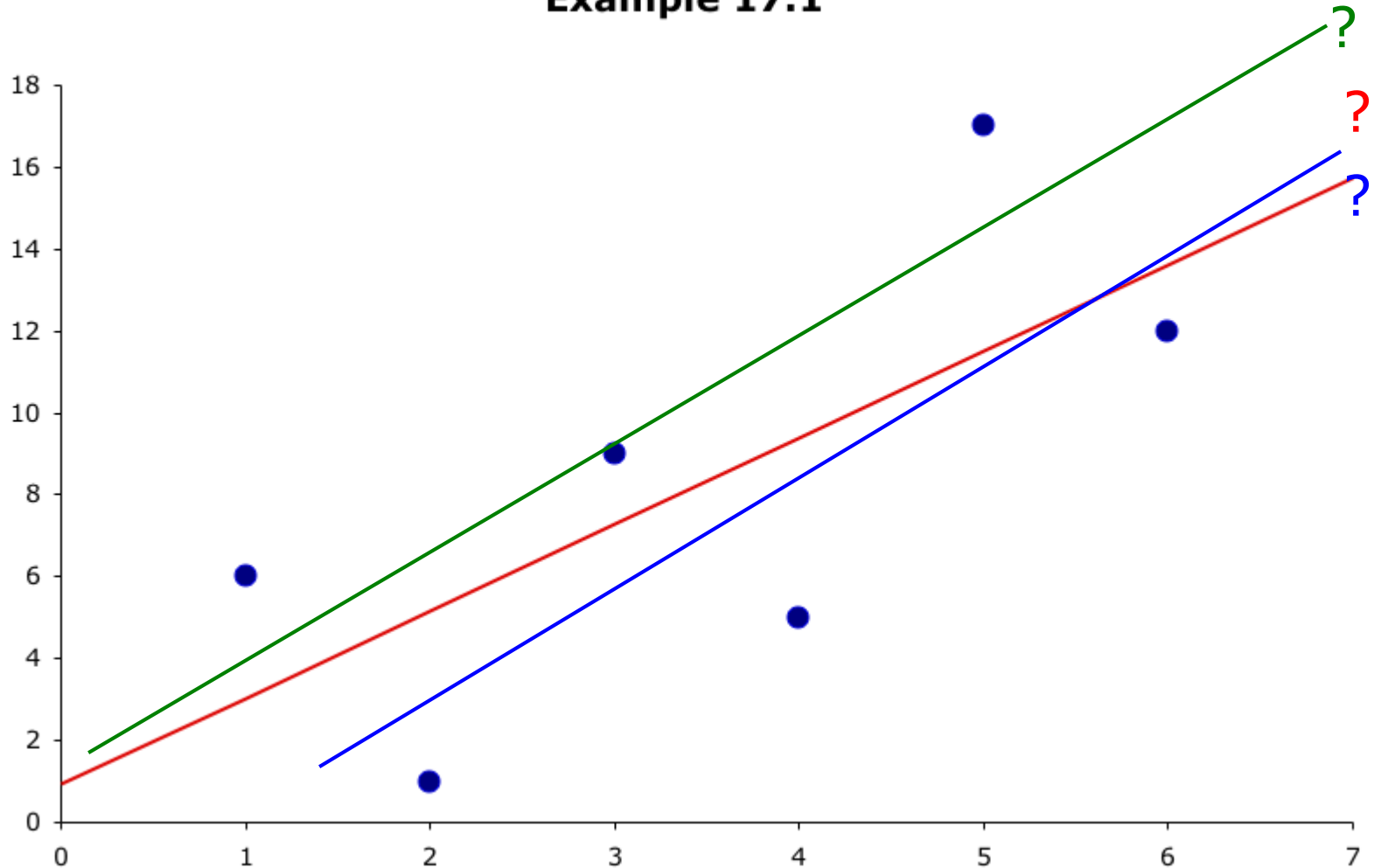
Simple Linear Regression Model...

- Note that both β_0 and β_1 are **population parameters** which are usually unknown and hence **estimated** from the data.



Which line has the best “fit” to the data?

Example 17.1



Estimating the Coefficients...

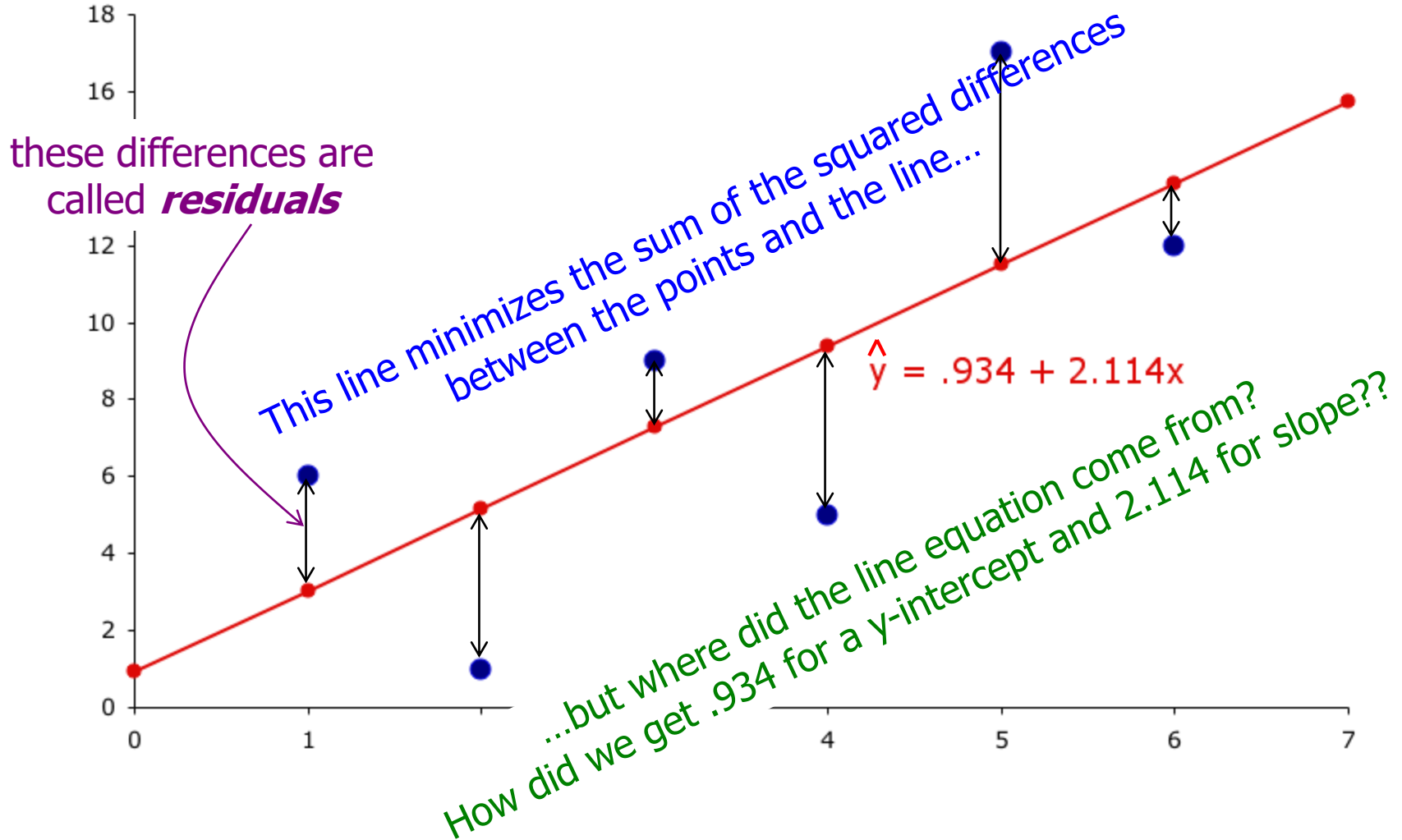
- In much the same way we based estimates of μ on $\bar{x}$, we estimate β_0 from b_0 and β_1 from b_1 , the y-intercept and slope (respectively) of the **least squares** or **regression line** given by:

$$\hat{y} = b_0 + b_1x$$

- (Recall: this is an application of the least squares method and it produces a straight line that **minimizes** the sum of the squared differences between the points and the line)

Least Squares Line...

Example 17.1



Mathematical Optimization Routine (Optional)

$$\min_{\{b_0, b_1\}} \left\{ \sum_i^n (y_i - \hat{y}_i)^2 \right\}$$

$$\Rightarrow \min_{\{b_0, b_1\}} \left\{ \sum_i^n (y_i - (b_0 + b_1 x_i))^2 \right\}$$

$$\Rightarrow \min_{\{b_0, b_1\}} \sum_i^n (y_i - b_0 - b_1 x_i)^2$$

Taking derivatives we get the first order conditions:

$$\frac{\partial}{\partial b_0} : (-1)2 \sum_i^n (y_i - b_0 - b_1 x_i) = 0,$$

$$\frac{\partial}{\partial b_1} : 2 \sum_i^n (y_i - b_0 - b_1 x_i)(-x_i) = 0.$$

Solving for b_0 and b_1 we get :

$$\sum_i^n y_i = \sum_i^n b_0 + b_1 \sum_i^n x_i, \quad (1)$$

$$\sum_i^n y_i x_i = b_0 \sum_i^n x_i + b_1 \sum_i^n x_i^2. \quad (2)$$

And further from (1) we get :

$$\sum_i^n y_i = nb_0 + b_1 \sum_i^n x_i \Rightarrow b_0 = \frac{1}{n} \sum_i^n y_i - \frac{b_1}{n} \sum_i^n x_i,$$

or

$$b_0 = \bar{y} - b_1 \bar{x}.$$

Substituting b_0 back in (2)

$$\sum_i^n y_i x_i = (\bar{y} - b_1 \bar{x}) \sum_i^n x_i + b_1 \sum_i^n x_i^2,$$

$$\Rightarrow \sum_i^n y_i x_i = \bar{y} \sum_i^n x_i - b_1 \bar{x} \sum_i^n x_i + b_1 \sum_i^n x_i^2,$$

$$\Rightarrow b_1 = \frac{\sum_i^n y_i x_i - \bar{y} \sum_i^n x_i}{\sum_i^n x_i^2 - \bar{x} \sum_i^n x_i} = \frac{\sum_i^n y_i x_i - \bar{y} n \bar{x}}{\sum_i^n x_i^2 - \bar{x} n \bar{x}},$$

$$\Rightarrow b_1 = \frac{\sum_i^n y_i x_i - n \bar{y} \bar{x}}{\sum_i^n x_i^2 - n \bar{x}^2} = \frac{\frac{1}{n} \sum_i^n y_i x_i - \bar{y} \bar{x}}{\frac{1}{n} \sum_i^n x_i^2 - \bar{x}^2}$$

$$\Rightarrow b_1 = \frac{s_{xy}}{s_x}.$$

Least Squares Line...

- The coefficients b_1 and b_0 for the least squares line...

$$\hat{y} = b_0 + b_1x$$

- ...are calculated as:

$$s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

$$s_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

$$b_1 = \frac{s_{xy}}{s_x^2}$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} \quad \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

Least Squares Line...

Recall...

Data

Statistics

Information

$$s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Data Points:

x	y
1	6
2	1
3	9
4	5
5	17
6	12

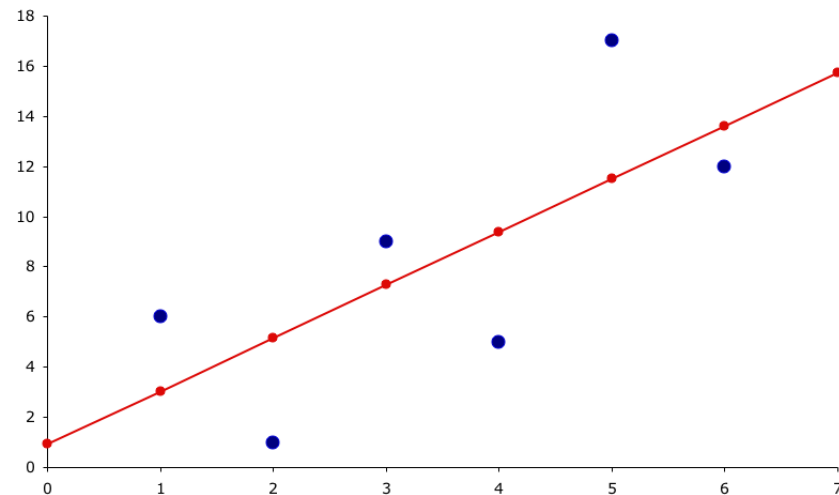
$$s_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

$$b_1 = \frac{s_{xy}}{s_x^2}$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} \quad \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Example 17.1

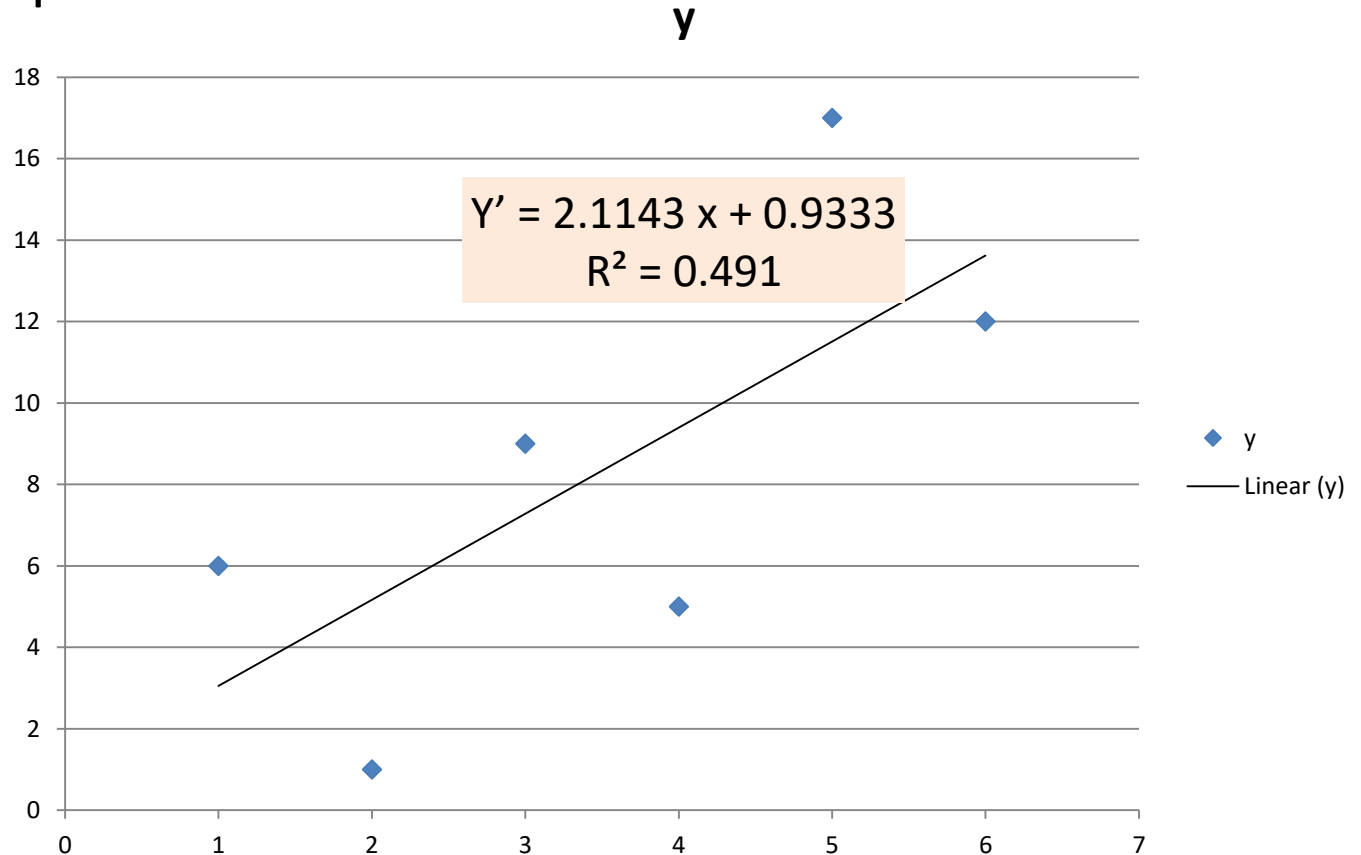


$$\hat{y} = .934 + 2.114x$$

Excel

- Insert – Scatter
- Right click on data point – add trend line (linear)
- Tick on add equation and add R^2

x	y
1	6
2	1
3	9
4	5
5	17
6	12



Annual Bonus and Years of Experience

Annual Bonus and Years of Experience

The annual bonuses (\$1,000s) of six employees with different years of experience were recorded as follows. We wish to determine the straight-line relationship between annual bonus and years of experience.

Years of experience x	1	2	3	4	5	6
Annual bonus y	6	1	9	5	17	12

Annual Bonus and Years of Experience

To apply the shortcut formula, we need to compute four summations. Using a calculator, we find

$$\sum_{i=1}^n x_i = 21$$

$$\sum_{i=1}^n y_i = 50$$

$$\sum_{i=1}^n x_i y_i = 212$$

$$\sum_{i=1}^n x_i^2 = 91$$

The covariance and the variance of x can now be computed:

$$s_{xy} = \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n} \right] = \frac{1}{6-1} \left[212 - \frac{(21)(50)}{6} \right] = 7.4$$

Annual Bonus and Years of Experience

$$s_x^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} \right] = \frac{1}{6-1} \left[91 - \frac{(21)^2}{6} \right] = 3.5$$

The sample slope coefficient is calculated next:

$$b_1 = \frac{s_{xy}}{s_x^2} = \frac{7.4}{3.5} = 2.114$$

The y -intercept is computed as follows:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{21}{6} = 3.5$$

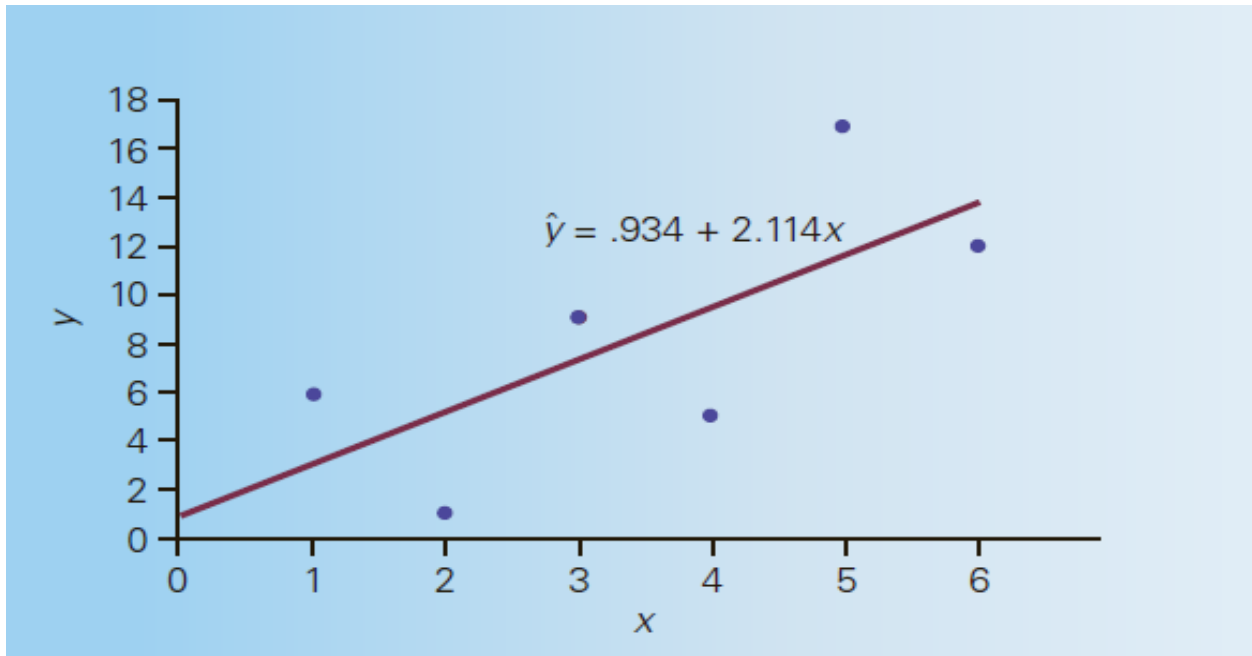
$$\bar{y} = \frac{\sum y_i}{n} = \frac{50}{6} = 8.333$$

$$b_0 = \bar{y} - b_1 \bar{x} = 8.333 - (2.114)(3.5) = .934$$

Annual Bonus and Years of Experience

Thus, the least squares line is

$$\hat{y} = .934 + 2.114x$$

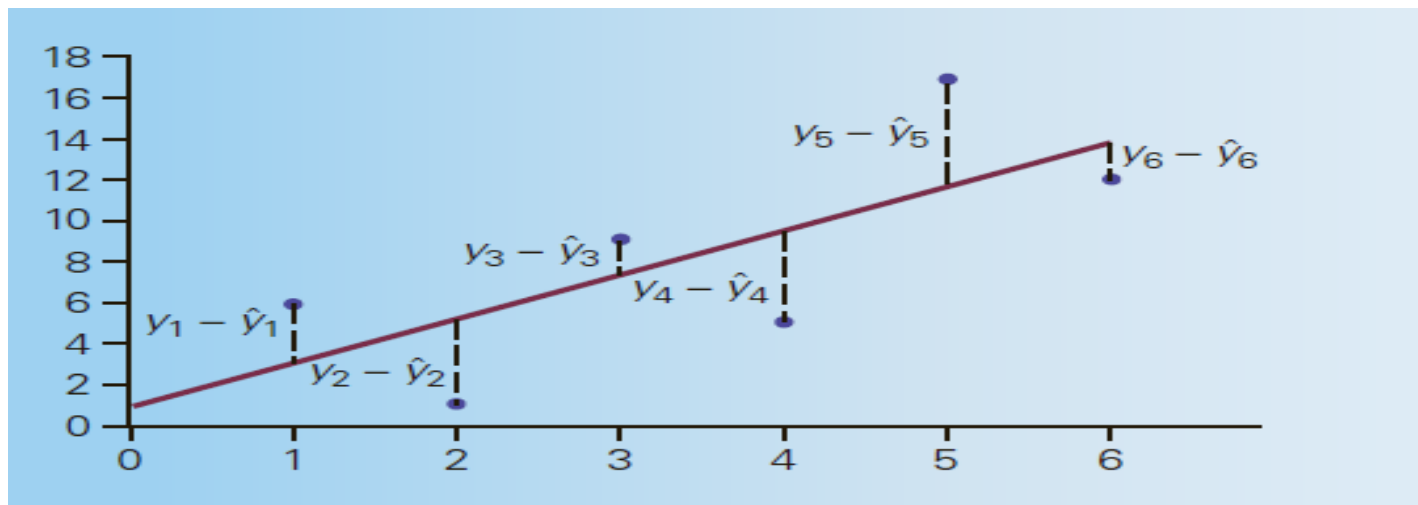


Annual Bonus and Years of Experience

- We can measure how well by computing the value of the minimized sum of squared deviations. The deviations between the actual data points and the line are called **residuals**, denoted e_i ; that is,

$$e_i = y_i - \hat{y}_i$$

- The residuals are observations of the error variable. Consequently, the minimized sum of squared deviations is called the **sum of squares for error**, denoted SSE.



Annual Bonus and Years of Experience

x_i	y_i	$\hat{y}_i = .934 + 2.114x_i$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$
1	6	3.048	2.952	8.714
2	1	5.162	-4.162	17.322
3	9	7.276	1.724	2.972
4	5	9.390	-4.390	19.272
5	17	11.504	5.496	30.206
6	12	13.618	-1.618	2.618
				$\sum (y_i - \hat{y}_i)^2 = 81.104$

Thus, $SSE = 81.104$. No other straight line will produce a sum of squared deviations as small as 81.104. In that sense, the regression line fits the data best. The sum of squares for error is an important statistic because it is the basis for other statistics that assess how well the linear model fits the data.

Least Square Line ...

Exercise : Decide which of these equations fits the given data best.

- MJT = Mean January Temperature
- LAT = Latitude

- A: $\text{MJT}' = 100 - 1.8 (\text{LAT})$
- B: $\text{MJT}' = 108 - 2.1 (\text{LAT})$
- C: $\text{MJT}' = 115 - 2.5 (\text{LAT})$

City	MJT	LAT
Boise	22	43.7
Philly	24	40.9
Milwaukee	13	43.3

Model predictions:

Model Comparison

<u>Lat (x)</u>	<u>A (y')</u>	<u>B (y')</u>	<u>C (y')</u>	<u>MJT (y)</u>
43.7	21.34	16.23	5.75	22
40.9	26.38	22.11	12.75	24
43.3	22.06	17.07	6.75	13

	<u>MJT-MJT_A</u>	<u>MJT-MJT_B</u>	<u>MJT-MJT_C</u>
	0.66	5.77	16.25
	-2.38	1.89	11.25
	-9.06	-4.07	6.25
Sum:	-10.78	3.59	33.75

	<u>(MJT-MJT_A)^2</u>	<u>(MJT-MJT_B)^2</u>	<u>(MJT-MJT_C)^2</u>
	0.4356	33.2929	264.0625
	5.6644	3.5721	126.5625
	82.0836	16.5649	39.0625
Sum^2:	88.1836	53.4299	429.6875

The Multiple Regression Model

The equation that describes how the dependent variable y is related to the independent variables $x_1, x_2, \dots, x_k$ and an error term. Idea: Examine the linear relationship between 1 dependent (Y) & 2 or more independent variables (X_i)

The diagram shows the equation $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$. Three labels in pink boxes are positioned above the equation with blue arrows pointing to specific terms: 'Y-intercept' points to β_0 , 'Population slopes' points to β_1 and β_k , and 'Random Error' points to ϵ .

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

Multiple Regression Model with k Independent Variables:

Multiple Regression Model with k Independent Variables

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_k X_{ki} + \varepsilon_i$$

where

β_0 = Y intercept

β_1 = slope of Y with variable X_1 , holding variables $X_2, X_3, \dots, X_k$ constant

β_2 = slope of Y with variable X_2 , holding variables $X_1, X_3, \dots, X_k$ constant

β_3 = slope of Y with variable X_3 , holding variables $X_1, X_2, \dots, X_k$ constant

$\vdots$

β_k = slope of Y with variable X_k holding variables $X_1, X_2, X_3, \dots, X_{k-1}$ constant

ε_i = random error in Y for observation i

MULTIPLE REGRESSION MODEL WITH TWO INDEPENDENT VARIABLES

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

β_0 represents the Y intercept,

- β_1 : the change in Y per unit change in X_1 , taking into account the effect of X_2 .
- β_2 : the change in Y per unit change in X_2 , taking into account the effect of X_1 .

Standard Multiple Regression Assumptions

- The values x_i and the error terms ε_i are independent
- The error terms are random variables with mean 0 and a constant variance, σ^2 .

$$E[\varepsilon_i] = 0 \quad \text{and} \quad E[\varepsilon_i^2] = \sigma^2 \quad \text{for } (i = 1, \dots, n)$$

(The constant variance property is called
homoscedasticity)

Standard Multiple Regression Assumptions

(continued)

- The random error terms, ε_i , are not correlated with one another, so that

$$E[\varepsilon_i \varepsilon_j] = 0 \quad \text{for all } i \neq j$$

- It is not possible to find a set of numbers, $c_0, c_1, \dots, c_k$, such that

$$c_0 + c_1 X_{1i} + c_2 X_{2i} + \dots + c_K X_{Ki} = 0$$

(This is the property of no linear relation for the X_j 's)

Multiple Regression Equation

The coefficients of the multiple regression model are estimated using sample data

Multiple regression equation with k independent variables:

The diagram shows the multiple regression equation $\hat{y}_i = b_0 + b_1x_{1i} + b_2x_{2i} + \dots + b_kx_{ki}$. Above the equation, three light blue boxes with black text provide labels. The first box, 'Estimated (or predicted) value of y', has a blue arrow pointing to $\hat{y}_i$. The second box, 'Estimated intercept', has a blue arrow pointing to b_0 . The third box, 'Estimated slope coefficients', has three blue arrows pointing to b_1 , b_2 , and b_k .

$$\hat{y}_i = b_0 + b_1x_{1i} + b_2x_{2i} + \dots + b_kx_{ki}$$

A simple random sample is used to compute sample statistics $b_0, b_1, b_2, \dots, b_k$ that are used as the point estimators of the parameters $\beta_0, \beta_1, \beta_2, \dots, \beta_k$.

Omni Power Bars

- In the Omni Power Bars scenario, your business objective, to determine the effect that price and in-store promotional expenses will have on sales, calls for examining a multiple regression model in which the price of an Omni Power bar in cents (X_1) and the monthly budget for in-store promotional expenses in dollars (X_2) are the independent variables and the number of Omni Power bars sold in a month (Y) is the dependent variable.
- To develop this model, you collect data from a sample of 34 stores in a supermarket chain selected for a test-market study of Omni Power. You choose stores in a way to ensure that they all have approximately the same monthly sales volume

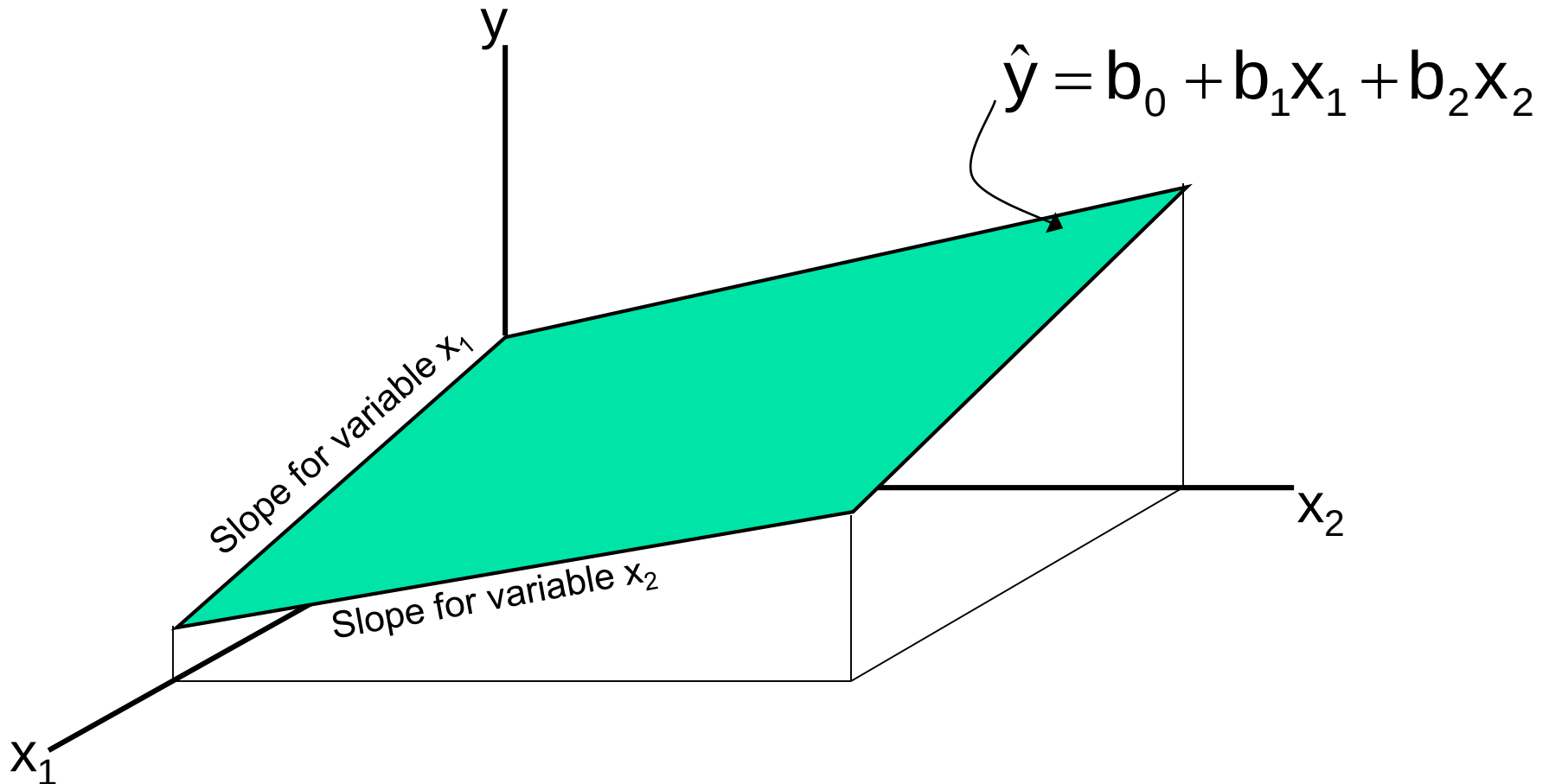
Omni Power Bars

Store	Sales	Price	Promotion	Store	Sales	Price	Promotion
1	4,141	59	200	18	2,730	79	400
2	3,842	59	200	19	2,618	79	400
3	3,056	59	200	20	4,421	79	400
4	3,519	59	200	21	4,113	79	600
5	4,226	59	400	22	3,746	79	600
6	4,630	59	400	23	3,532	79	600
7	3,507	59	400	24	3,825	79	600
8	3,754	59	400	25	1,096	99	200
9	5,000	59	600	26	761	99	200
10	5,120	59	600	27	2,088	99	200
11	4,011	59	600	28	820	99	200
12	5,015	59	600	29	2,114	99	400
13	1,916	79	200	30	1,882	99	400
14	675	79	200	31	2,159	99	400
15	3,636	79	200	32	1,602	99	400
16	3,224	79	200	33	3,354	99	600
17	2,295	79	400	34	2,927	99	600

Multiple Regression Equation

(continued)

Two variable model



Least Squares Method

Least Squares Criterion

$$\min \sum (y_i - \hat{y}_i)^2$$

Computation of Coefficient Values

The formulas for the regression coefficients $b_0, b_1, b_2, \dots, b_k$ involve the use of matrix algebra.

Least Squares Method

The basic linear multiple regression model is

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \cdots + \beta_K X_{Ki} + \epsilon_i$$

The n -observations on the dependent variable and the K independent variables can be represented as follows:

$$Y_1 = \beta_0 + \beta_1 X_{11} + \beta_2 X_{21} + \beta_3 X_{31} + \cdots + \beta_K X_{K1} + \epsilon_1$$

$$Y_2 = \beta_0 + \beta_1 X_{12} + \beta_2 X_{22} + \beta_3 X_{32} + \cdots + \beta_K X_{K2} + \epsilon_2$$

$$Y_3 = \beta_0 + \beta_1 X_{13} + \beta_2 X_{23} + \beta_3 X_{33} + \cdots + \beta_K X_{K3} + \epsilon_3$$

.....

.....

.....

$$Y_n = \beta_0 + \beta_1 X_{1n} + \beta_2 X_{2n} + \beta_3 X_{3n} + \cdots + \beta_K X_{Kn} + \epsilon_n$$

Least Squares Method

This appears in matrix form as

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & X_{21} & X_{31} & \cdots & \cdots & X_{K1} \\ 1 & X_{12} & X_{22} & X_{32} & \cdots & \cdots & X_{K2} \\ 1 & X_{13} & X_{23} & X_{33} & \cdots & \cdots & X_{K3} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & X_{1n} & X_{2n} & X_{3n} & \cdots & \cdots & X_{Kn} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \vdots \\ \beta_K \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{\mathcal{E}}$$

where $\mathbf{Y}$ is an n by 1 column vector, $\mathbf{X}$ is an n by $K + 1$ matrix (i.e., a matrix with n rows and $K + 1$ columns), $\mathbf{B}$ is a $K + 1$ by 1 column vector and $\mathbf{\mathcal{E}}$ is an n by 1 column vector. The first column of the matrix $\mathbf{X}$ is a column of 1's.

Least Squares Method

- Our problem is now to choose an estimate $\mathbf{b}$ such that

$$\mathbf{Y} = \mathbf{X}\mathbf{b} + \mathbf{e}$$

where $\mathbf{b}$ is a $K + 1$ by 1 column vector of point estimates of the vector $\mathcal{B}$ and $\mathbf{e}$ is an n by 1 column vector of residuals. According to the method of least squares we choose the vector $\mathbf{b}$ so as to minimize the sum of squared residuals which appears in matrix form as

$$\mathbf{e}'\mathbf{e} = \sum^n e_i^2,$$

$$\begin{aligned}\mathbf{e}'\mathbf{e} &= (\mathbf{Y} - \mathbf{X}\mathbf{b})'(\mathbf{Y} - \mathbf{X}\mathbf{b}) \\ &= (\mathbf{Y}' - \mathbf{b}'\mathbf{X}')(\mathbf{Y} - \mathbf{X}\mathbf{b}) \\ &= (\mathbf{Y}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}\mathbf{b} - \mathbf{b}'\mathbf{X}'\mathbf{Y} + \mathbf{b}'\mathbf{X}'\mathbf{X}\mathbf{b}) \\ &= (\mathbf{Y}'\mathbf{Y} - 2\mathbf{Y}'\mathbf{X}\mathbf{b} + \mathbf{b}'\mathbf{X}'\mathbf{X}\mathbf{b})\end{aligned}$$

Least Squares Method

- Normal Equation

$$\mathbf{X}'\mathbf{X}\hat{\mathbf{b}} = \mathbf{X}'\mathbf{Y}$$

- For K=3, we have

$$\begin{array}{rclclclcl} n \hat{b}_0 & + & \hat{b}_1 \sum X_{1i} & + & \hat{b}_2 \sum X_{2i} & = & \sum Y_i \\ \hat{b}_0 \sum X_{1i} & + & \hat{b}_1 \sum X_{1i}^2 & + & \hat{b}_2 \sum X_{1i}X_{2i} & = & \sum X_{1i}Y_i \\ \hat{b}_0 \sum X_{2i} & + & \hat{b}_1 \sum X_{1i}X_{2i} & + & \hat{b}_2 \sum X_{2i}^2 & = & \sum X_{2i}Y_i \end{array}$$

- After solving we get

$$\hat{\mathbf{b}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

where $(\mathbf{X}'\mathbf{X})^{-1}$ is the inverse of the matrix $\mathbf{X}'\mathbf{X}$.

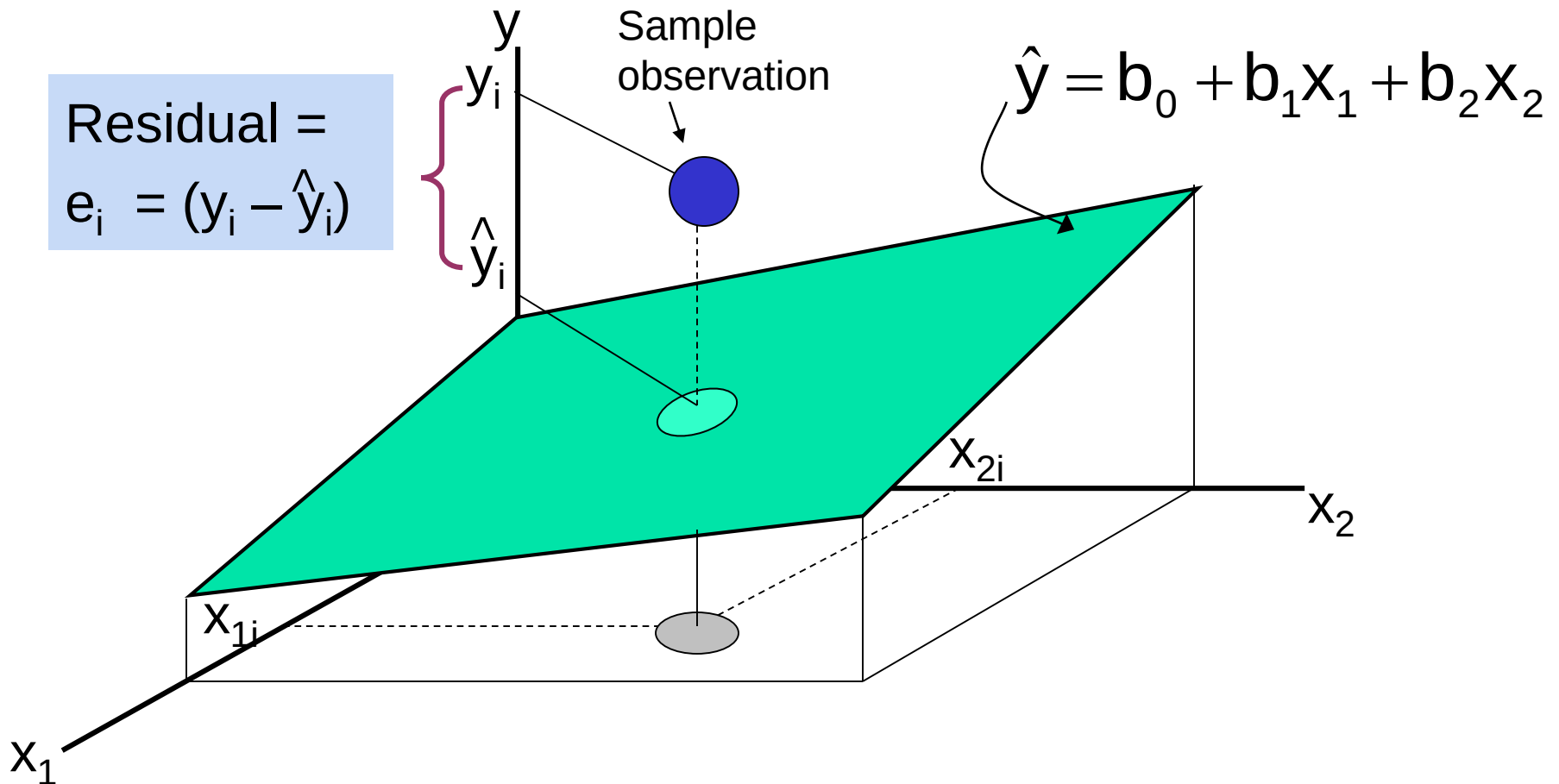
Omni Power Bars

	A	B	C	D	E	F	G
1	Regression Analysis						
2							
3	Regression Statistics						
4	Multiple R	0.8705					
5	R Square	0.7577					
6	Adjusted R Square	0.7421					
7	Standard Error	638.0653					
8	Observations	34					
9							
10	ANOVA						
11		<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
12	Regression	2	39472730.7730	19736365.3865	48.4771	0.0000	
13	Residual	31	12620946.6682	407127.3119			
14	Total	33	52093677.4412				
15							
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
17	Intercept	5837.5208	628.1502	9.2932	0.0000	4556.3999	7118.6416
18	Price	-53.2173	6.8522	-7.7664	0.0000	-67.1925	-39.2421
19	Promotional Expenses	3.6131	0.6852	5.2728	0.0000	2.2155	5.0106

Residuals in Multiple Regression

Two variable model

Residual =
 $e_i = (y_i - \hat{y}_i)$



Omni Power Bars

- The computed values of the net regression coefficients are

$$b_0 = 5,837.5208 \quad b_1 = -53.2173 \quad b_2 = 3.6131$$

Therefore, the multiple regression equation is

$$\hat{Y}_i = 5,837.5208 - 53.2173X_{1i} + 3.6131X_{2i}$$

where

$\hat{Y}_i$ = predicted monthly sales of OmniPower bars for store i

X_{1i} = price of OmniPower bar (in cents) for store i

X_{2i} = monthly in-store promotional expenses (in \$) for store i

Interpretation Regression coefficients

- The sample Y intercept, b_0 , estimates the number of Omni Power bars sold in a month if the price was zero cents and the total amount spent on promotional expenses was \$0.00.
- Using the net regression coefficients b_1 and b_2 , the effects of adding one cent to the price of Omni Power bars (X_1) or adding \$1 to monthly promotion expenditures (X_2) can be summarized.

Interpretation Regression coefficients

- **A price increase of one cent** - Predict mean Omni Power monthly sales to decrease by 53.2173 bars, holding constant the promotional expenditures.
- **An increase of \$1 in monthly promotional expenditures** - Predict mean Omni Power monthly sales to increase by 3.6131 bars, holding constant the price.
- Which explains the effect of changing one independent variable while holding the value of the all other independent variables constant

Predicting the Dependent Variable Y

- What are the predicted mean sales for a store charging 79 cents during a month in which promotional expenses are \$400?

$$\hat{Y}_i = 5,837.5208 - 53.2173X_{1i} + 3.6131X_{2i}$$

with $X_{1i} = 79$ and $X_{2i} = 400$,

$$\begin{aligned}\hat{Y}_i &= 5,837.5208 - 53.2173(79) + 3.6131(400) \\ &= 3,078.57\end{aligned}$$

- Thus, you predict that stores charging 79 cents and spending \$400 in promotional expenses will sell a mean of 3,078.57 Omni Power bars per month.

Multiple Regression Assumptions

Errors (residuals) from the regression model:

$$e_i = (y_i - \hat{y}_i)$$

Assumptions:

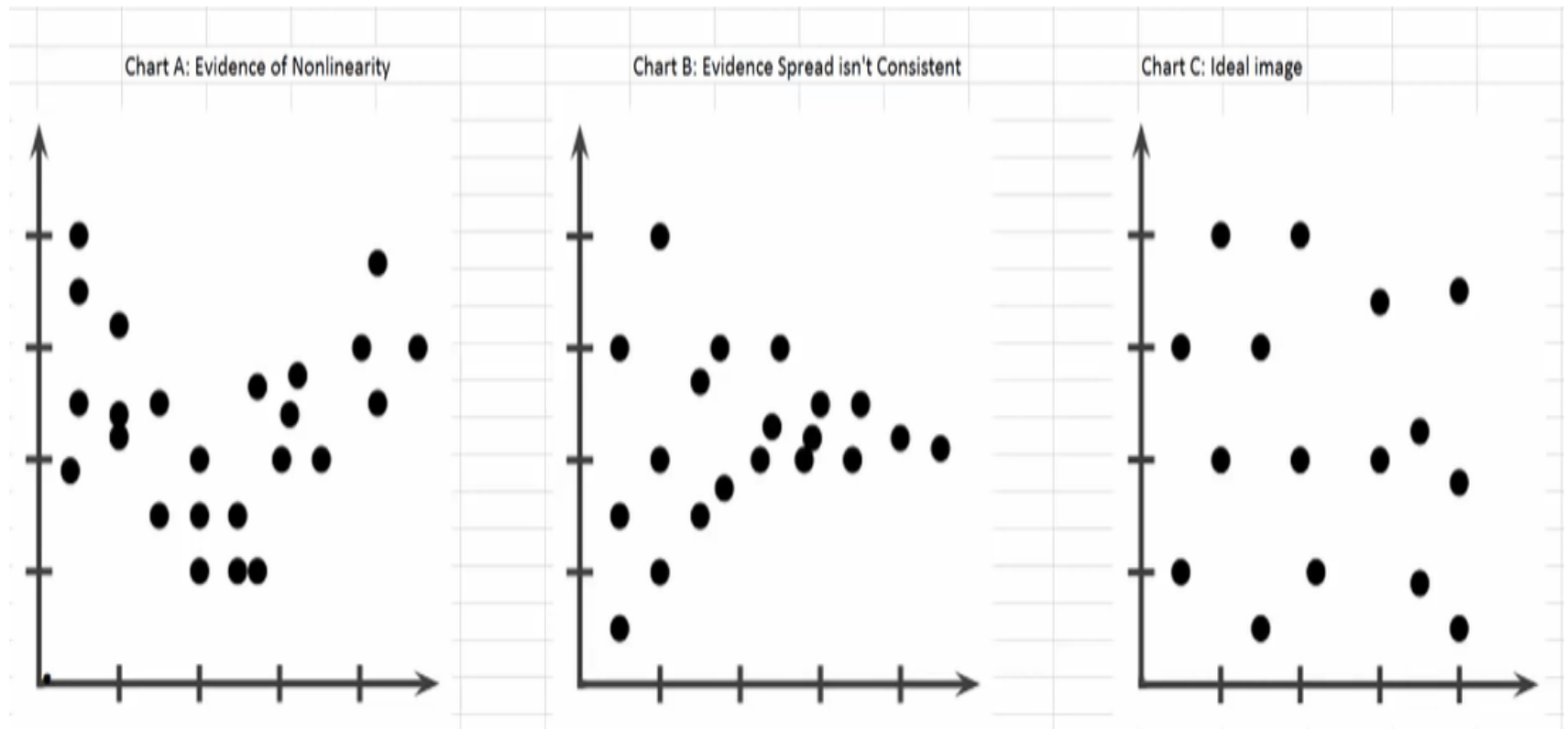
- The errors are normally distributed
- Errors have a constant variance
- The model errors are independent

Analysis of Residuals in Multiple Regression

- These residual plots are used in multiple regression:
 - Residuals vs. $\hat{y}_i$
 - Residuals vs. x_{1i}
 - Residuals vs. x_{2i}
 - Residuals vs. time (if time series data)

Use the residual plots to check for violations of regression assumptions

Interpreting Multiple Regression Residual Plots



Interpreting Multiple Regression Residual Plots

- **Residuals versus the predicted value of Y**

Evidence of a possible curvilinear effect in at least one independent variable, a possible violation of the assumption of equal variance, and/or the need to transform the Y variable

- **Residuals versus an independent X variable**

Suggests evidence of a curvilinear effect and, therefore, indicates the need to add a curvilinear independent variable to the multiple regression model

How Well Does the Model Fit?

- We would like Y and $\hat{Y}$ to be as close to one another as possible (or the *difference* between Y and $\hat{Y}$ to be as *small* as possible).
- In Fact, we want ALL the differences-squared ADDED UP to be as small as possible.
- The sum of all total differences are called the Sum of Squared Errors or **SSE**.
 - a. $(Y - \hat{Y}) = e$, the estimated error, or residual.
 - b.
$$SSE = \sum_i (Y_i - \hat{Y}_i)^2 = \sum_i e_i^2$$
 - c. Aggregate measure of how much the predicted Y 's differ from the actual Y 's.
 - d. The **bs** minimizing SSE construct the Least Square Line.

How Well Does the Model Fit?

$$\mathbf{TSS = SSR + SSE}$$

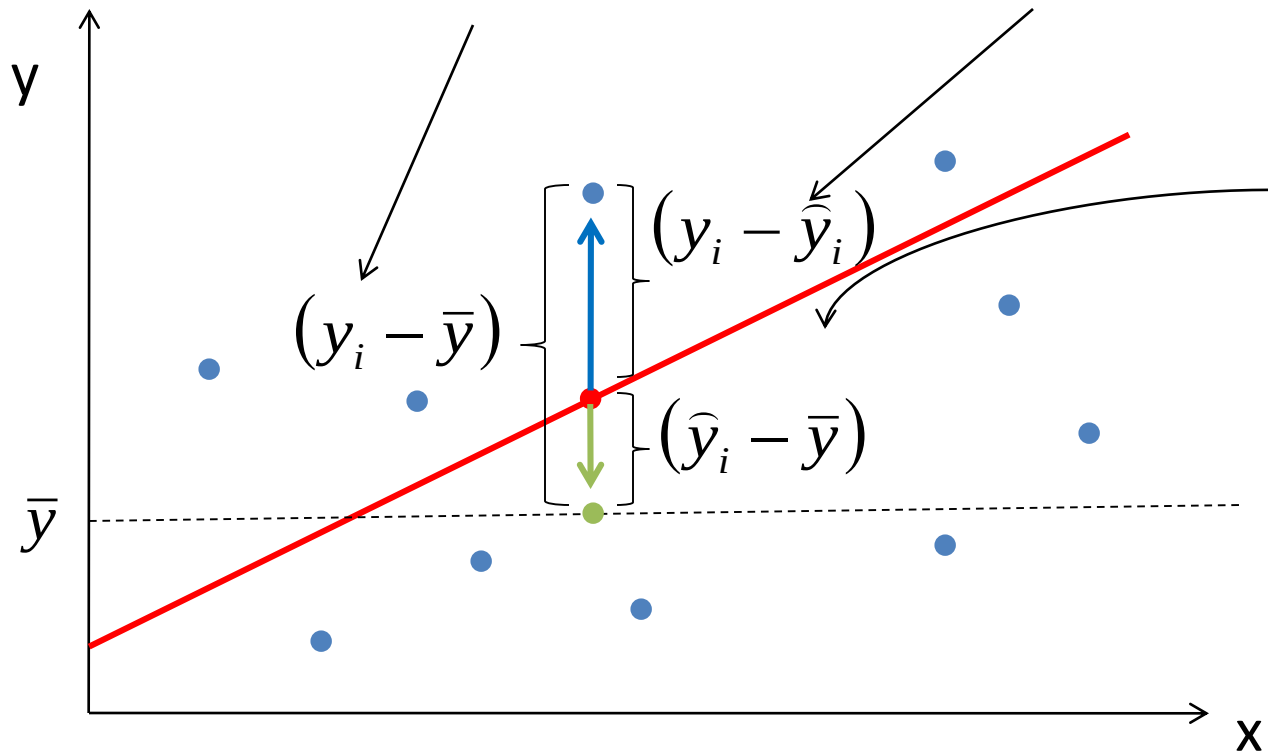
- The total variation (Total Sum of Squares: **TSS**) of the dependent variable can be divided into:
 - The variation explained by the model (Sum of Squares for Regression: **SSR**) and
 - The variation unexplained (Sum of Squared Error: **SSE**).
- We measure the variation around the mean, $\bar{Y}$.

$$TSS = \sum_i (Y_i - \bar{Y})^2, \text{ total variation around the mean}$$

$$SSR = \sum_i (\hat{Y}_i - \bar{Y})^2, \text{ explained variation around the mean}$$

$$SSE = \sum_i (Y_i - \hat{Y}_i)^2, \text{ unexplained variation around the mean}$$

$$TSS = SSR + SSE$$



$$TSS = \sum_i (Y_i - \bar{Y})^2, \quad SSR = \sum_i (\hat{Y}_i - \bar{Y})^2,$$

$$SSE = \sum_i (Y_i - \hat{Y}_i)^2$$

Coefficient of Determination, R^2

- Reports the proportion of total variation in y explained by all x variables taken together

$$R^2 = \frac{SSR}{TSS} = \frac{\text{regression sum of squares}}{\text{total sum of squares}}$$

- This is the ratio of the explained variability to total sample variability

Coefficient of Determination, R^2

$$R^2 = \frac{\sum_i (\hat{Y}_i - \bar{Y})^2}{\sum_i (Y_i - \bar{Y})^2} = \frac{SSR}{TSS}, \text{ or}$$

$$R^2 = 1 - \frac{\sum_i (Y_i - \hat{Y}_i)^2}{\sum_i (Y_i - \bar{Y})^2} = 1 - \frac{SSE}{TSS}$$

- **SSE** - Sum of Squared **E**rror
- **SSR** - Sum of Squares for **R**egression
- **TSS** – **T**otal Sum of Squares

Coefficient of Determination, R^2

1. The proportion of the deviation of Y from its mean in the sample that is explained by the regression.
2. It is always between 0 and 1.
3. The closer to 1, the better is the fit.
4. It happens also to be the correlation squared:

$$R^2 = \rho^2$$

where ρ is the correlation coefficient

5. It is the most commonly used measure of the “fit” of a regression.

Coefficient of Determination, R^2

- In the Omni Power example, $SSR = 39,472,730.77$ and $TSS = 52,093,677.44$.
- $$R^2 = \frac{SSR}{TSS} = \frac{39,472,730.77}{52,093,677.44} = 0.7577$$
- The coefficient of multiple determination, $R^2 = 0.7577$, indicates that 75.77% of the variation in sales is explained by the variation in the price and in the promotional expenses.

Estimation of Error Variance

- Consider the population regression model

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_K x_{Ki} + \varepsilon_i$$

- The unbiased estimate of the variance of the errors is

$$s_e^2 = \frac{\sum_{i=1}^n e_i^2}{n-K-1} = \frac{SSE}{n-K-1}$$

where $e_i = y_i - \hat{y}_i$

- The square root of the variance, s_e , is called the **standard error of the estimate**

Adjusted Coefficient of Determination, $\bar{R}^2$

- R^2 never decreases when a new X variable is added to the model, even if the new variable is not an important predictor variable
 - This can be a disadvantage when comparing models
- What is the net effect of adding a new variable?
 - We lose a degree of freedom when a new X variable is added
 - Did the new X variable add enough explanatory power to offset the loss of one degree of freedom?

Adjusted Coefficient of Determination, $\bar{R}^2$

(continued)

- Used to correct for the fact that adding non-relevant independent variables will still reduce the error sum of squares

$$\bar{R}^2 = 1 - \frac{\text{SSE} / (n - K - 1)}{\text{TSS} / (n - 1)}$$

(where n = sample size, K = number of independent variables)

- Adjusted R^2 provides a better comparison between multiple regression models with different numbers of independent variables
- Penalize excessive use of unimportant independent variables
- Smaller than R^2

Adjusted Coefficient of Determination, $\bar{R}^2$

- Alternative

$$R_{adj}^2 = 1 - \left[(1 - R^2) \frac{n - 1}{n - k - 1} \right]$$

For the OmniPower sales data, because $r^2 = 0.7577$, $n = 34$, and $k = 2$,

$$\begin{aligned} r_{adj}^2 &= 1 - \left[(1 - 0.7577) \frac{34 - 1}{34 - 2 - 1} \right] = 1 - \left[(0.2423) \frac{33}{31} \right] \\ &= 1 - 0.2579 \\ &= 0.7421 \end{aligned}$$

- Therefore, 74.21% of the variation in sales is explained by the multiple regression model adjusted for the number of independent variables and sample size.

Evaluating Individual Regression Coefficients

- Use t-tests for individual coefficients
- Shows if a specific independent variable is conditionally important
- Hypotheses:

- $H_0: \beta_j = 0$ (no linear relationship)
- $H_1: \beta_j \neq 0$ (linear relationship does exist between x_j and y)

Tests of Hypotheses for the Regression Coefficients

- If the regression errors, ε_i , are normally distributed and the standard regression assumptions hold, then the following hypothesis tests have significance level α :

1. To test either null hypothesis

$$H_0: \beta_j = \beta^* \quad \text{or} \quad H_0: \beta_j \leq \beta^*$$

against the alternative

$$H_1: \beta_j > \beta^*$$

the decision rule is as follows:

$$\text{reject } H_0 \text{ if } \frac{b_j - \beta^*}{s_{b_j}} > t_{n-K-1, \alpha}$$

Tests of Hypotheses for the Regression Coefficients

2. To test either null hypothesis

$$H_0: \beta_j = \beta^* \quad \text{or} \quad H_0: \beta_j \geq \beta^*$$

against the alternative

$$H_1: \beta_j < \beta^*$$

the decision rule is as follows:

$$\text{reject } H_0 \text{ if } \frac{b_j - \beta^*}{s_{b_j}} < -t_{n-K-1, \alpha}$$

3. To test the null hypothesis

$$H_0: \beta_j = \beta^*$$

against the two-sided alternative

$$H_1: \beta_j \neq \beta^*$$

the decision rule is as follows:

$$\text{reject } H_0 \text{ if } \frac{b_j - \beta^*}{s_{b_j}} > t_{n-K-1, \alpha/2} \quad \text{or} \quad \frac{b_j - \beta^*}{s_{b_j}} < -t_{n-K-1, \alpha/2}$$

Omni Power Bars

- To determine whether variable X_2 (amount of promotional expenses) has a significant effect on sales, after taking into account the effect of the price of Omni Power bars, the null and alternative hypotheses are

$$H_0: \beta_2 = 0$$

$$H_1: \beta_2 \neq 0$$

- Assuming H_0 is true

$$t_{STAT} = \frac{b_2 - \beta_2}{S_{b_2}}$$

$$= \frac{3.6131 - 0}{0.6852} = 5.2728$$

- If you select a level of significance of 0.05, Since it is a two-tailed test the critical values of t for 31 degrees of freedom from are -

Omni Power Bars

Result	Conclusions
$t_{STAT} = 5.2728$ is greater than 2.0395 $p\text{-value} = 0.0000$ is less than the level of significance, $\alpha = 0.05$	1. Reject the null hypothesis H_0. 2. Conclude that strong evidence exists for claiming that promotional expenses is related to the dependent Y variable, sales, taking into account the price. 3. The probability is 0.0000 that $t_{STAT} 5.2728$ $t_{STAT} < -5.2728$.

- At the 0.05 level of significance, is there evidence that the slope of sales with price is different from zero?

Confidence Interval Estimation

CONFIDENCE INTERVAL ESTIMATE FOR THE SLOPE

$$b_j \pm t_{\alpha/2} S_{b_j}$$

where

$t_{\alpha/2}$ = the critical value corresponding to an upper-tail probability of $\alpha/2$
(a cumulative area of $1 - \alpha/2$) from the t distribution with $n - k - 1$
degrees of freedom

k = the number of independent variables

To construct a 95% confidence interval estimate of the population slope, b_1 (the effect of price, X_1 , on sales, Y , holding constant the effect of promotional expenses, X_2), the critical value of t at the 95% confidence level with 31 degrees of freedom is 2.0395.

Confidence Interval Estimation

$$\begin{aligned}b_1 &\pm t_{\alpha/2} S_{b_1} \\-53.2173 &\pm (2.0395)(6.8522) \\-53.2173 &\pm 13.9752 \\-67.1925 &\leq \beta_1 \leq -39.2421\end{aligned}$$

- Taking into account the effect of promotional expenses, the estimated effect of a 1-cent increase in price is to reduce mean sales by approximately 39.2 to 67.2 bars. You have 95% confidence that this interval correctly estimates the relationship between these variables. From a hypothesis-testing viewpoint, because this confidence interval does not include 0, you conclude that the regression coefficient, for price, has a significant effect.

Confidence Interval Estimation

- Construct a 95% confidence interval estimate of the population slope of sales with promotional expenses.

$$b_2 \pm t_{\alpha/2} S_{b_2}$$

$$3.6131 \pm (2.0395)(0.6852)$$

$$3.6131 \pm 1.3975$$

$$2.2156 \leq \beta_2 \leq 5.0106$$

Thus, taking into account the effect of price, the estimated effect of each additional dollar of promotional expenses is to increase mean sales by approximately 2.22 to 5.01 bars. You have 95% confidence that this interval correctly estimates the relationship between these variables. From a hypothesis-testing viewpoint, because this confidence interval does not include 0, you can conclude that the regression coefficient, β_2 , has a significant effect.

Test on All Coefficients

- F-Test for Overall Significance of the Model
- Shows if there is a linear relationship between all of the X variables considered together and Y
- Use F test statistic
- Hypotheses:

$H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$ (no linear relationship)

$H_1: \text{at least one } \beta_i \neq 0$ (at least one independent variable affects Y)

ANOVA Summary Table for the Overall F Test

Source of Variation	SS	df	MS (Variance)	F ratio
Regression	SSR	K	$MSR = \frac{SSR}{K}$	$F = \frac{MSR}{MSE}$
Error	SSE	$n - K - 1$	$MSE = \frac{SSE}{n - K - 1}$	
Total	$TSS = SSR + SSE$	$n - 1$		

F-Test for Overall Significance

- Test statistic:

$$F = \frac{MSR}{s_e^2} = \frac{SSR/K}{SSE/(n-K-1)}$$

where F has k (numerator) and
 $(n - K - 1)$ (denominator)
degrees of freedom

- The decision rule is

$$\text{Reject } H_0 \text{ if } F > F_{k, n-K-1, \alpha}$$

Omni Power Bars

- For the Omni Power sales study, using a level of significance, $\alpha = 0.05$, the critical value of the F distribution with 2 and 31 degrees of freedom is approximately 3.32.
- **Results:** $F_{STAT} = 48.4771$ is greater than the F critical value, 3.32, p -value = 0.0000 is less than the level of significance, $\alpha = 0.05$
- **Conclusions:**
 1. Reject the null hypothesis H_0 .
 2. Conclude that evidence exists for claiming that at least one of the independent X variables (price or promotional expenses) is related to the dependent Y variable, sales.
 3. The probability is 0.0000 that $F_{STAT} > 48.4771$.